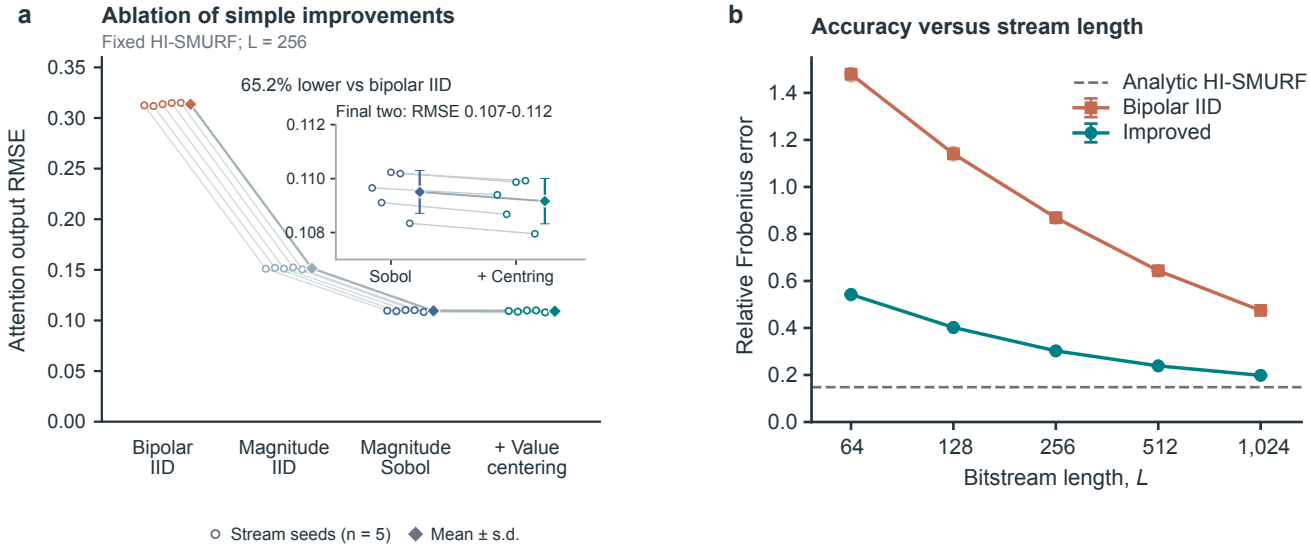

a
Ablation of simple improvements
Fixed HI-SMURF; L = 256
65.2% lower vs bipolar IID
Final two: RMSE 0.107-0.112
Attention output RMSE
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0.112
0.110
0.108
Sobol
+ Centring
Bipolar IID
Magnitude IID
Magnitude Sobol
+ Value centering
Stream seeds (n = 5)
Mean ± s.d.
b
Accuracy versus stream length
Analytic HI-SMURF
Bipolar IID
Improved
Relative Frobenius error
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
64
128
256
512
1,024
Bitstream length, L